

Question Paper

Exam Date & Time: 30-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Generative AI and Applications [IT_4451]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) You are a research engineer developing a generative AI prototype to create short music snippets based on text prompts (e.g., "generate a calm piano tune"). The team is unsure which model architecture to begin with. (5)
- A) Construct a **plan** to design this generative AI system by specifying:
1. The **type of data** and **model architecture** to use (RNN, Transformer, or GAN-based).
 2. The **training approach** (supervised, unsupervised, or hybrid).

The **evaluation metric(s)** suitable for generated outputs.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) **Illustrate** how a generative pretrained model uses stacked transformer decoder blocks to compute the probability of each token in a sequence. (3)
- Write the corresponding equations for:
1. Input representation h_0 ,
 2. Transformer block computation h_i , and
- Output probability $P(x_i)$ using the softmax function.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) In your own words, briefly explain and provide examples of how Generative Pre-trained Transformers (GPT) are being utilized across three distinct professional fields, such as software development, creative writing, or customer service. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2) Explain the different types of Recurrent Neural Networks (RNNs) based on input-output cardinality (one-to-one, one-to-many, many-to-one, and many-to-many) with suitable real-world examples for each. Discuss how these architectures differ in handling sequential data and where each type is best applied. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) You are designing a language translation model for an Indian startup that aims to translate English text into French. The team suggests using an RNN-based approach, but you propose using a Transformer model instead. (3)

Apply your understanding of the Transformer encoder-decoder architecture to justify your choice over RNN.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Identify the major technological milestones in the evolution of Generative AI from early symbolic models to modern multimodal systems. Mention key breakthroughs such as VAEs, GANs, Transformers, Diffusion Models, and recent multimodal advancements. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3) Develop and analyze the mathematical model of a Denoising Autoencoder (DAE). Explain how noise is introduced to the input data, how the model learns to reconstruct the original data through the encoder and decoder functions, and discuss its role in feature learning and robust representation. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) A decoder-based language model assigns the following probabilities to a 5-token sequence: (3)

$$P = [0.6, 0.1, 0.1, 0.1, 0.1]$$

Using the perplexity formula

$$PP = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i)}$$

Calculate the model's perplexity. Explain what it means if the perplexity is higher or lower.

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Describe the concept of Autoregressive Generation in GPT models and mention any two real-world applications where GPT is commonly used. (2)

Course Outcome: "CO5" / Cognitive Level: "Remember"

- 4) Apply your understanding to develop an explanation of how pretraining and fine-tuning stages build and enhance large language models' effectiveness. Use examples to illustrate how pretraining large-scale data develops general linguistic knowledge and how fine-tuning adapts the model to specific tasks such as question answering or summarization. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Construct and analyze the working principle of Variational Autoencoders (VAEs). Explain their mathematical formulation involving the encoder, decoder, and latent variable learning, and discuss how VAEs differ from traditional autoencoders in generating new data samples. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Interpret how the use of real-world datasets can lead to privacy and security concerns in large language model training. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 5) Construct and analyze the architecture of a Bidirectional Recurrent Neural Network (BiRNN). Explain how processing sequences in both forward and backward directions improves context understanding and discuss its advantages over a standard RNN with suitable examples. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) While designing a Transformer-based language generator, your model begins producing grammatically incorrect sentences because it is using information from words that haven't been generated yet. (3)
- Apply your understanding of the masked self-attention mechanism to explain how it prevents the

model from accessing future tokens during decoding.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Describe the main purpose of an **LSTM (Long Short-Term Memory)** network and explain how it overcomes the limitations of a traditional RNN. (2)

Course Outcome: "CO2" / Cognitive Level: "Remember"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_4451:Generative AI and Applications

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	IT_4451:Generative AI and Applications

#	Level	No of Questions	Marks (%)
1	Remember	2	4(8%)
2	Understand	3	6(12%)
3	Application	5	15(30%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)